

M3Jar — Phase 4C Closure Report

Finalize Phase 4C: Contributor Fund Commitment and Principal Refunds

Generated: 2026-08-05 | Commit: b6a4694 | Branch: main | Remote: github.com/JB102298/TripJar.git

1. Scope & Constraints

This report documents the correction-and-closure pass applied to the Phase 4C implementation (commit 52f1eab). All corrections were limited to the items identified in the pre-correction conformance audit.

- No Phase 4D work included.
- Stripe TEST MODE only — no live credentials, no real charges.
- No production scheduler wired (noted as pre-go-live prerequisite in code comment).
- No PAT used — push performed via Replit managed GitHub integration.
- No production DB, DNS, or deployment changes.

2. Corrections Applied

2a. Snapshot TTL: 5 min !' 15 min

File: artifacts/api-server/src/routes/fund-commitment.ts

The SNAPSHOT_TTL_MS constant was corrected from 5 minutes (300 000 ms) to 15 minutes (900 000 ms) to match the architecture specification.

2b. Mobile refund button visibility

File: artifacts/mobile/app/jar/[id].tsx

showRefundBtn previously required cutoffReached && phase === 'Saving'. The API allows refunds throughout the entire Saving phase (pre- and post-cutoff), so the cutoffReached guard was removed. showCommitBtn remains cutoff-gated.

Before: showRefundBtn = cutoffReached && phase === 'Saving'

After: showRefundBtn = phase === 'Saving' || phase === 'Commitment'

2c. Concurrent duplicate-commit race (HTTP 500 !' idempotent 2xx)

File: artifacts/api-server/src/routes/fund-commitment.ts

A race where two simultaneous confirms both passed the pre-tx idempotency check caused the second request to see 0 refundable balance and return HTTP 500. Fix: added an under-lock SELECT on fund_commitments inside the SELECT FOR UPDATE transaction. If a row already exists, the transaction exits cleanly (no-op commit, no ledger changes) and a typed concurrentWinner value is passed back to the handler, which then returns idempotent 200 (same snapshot) or 409 AlreadyCommitted (different snapshot). A TypeScript const-cast pattern was used to preserve type narrowing across the async closure boundary.

2d. Balance invariant documentation (4-term formula)

File: artifacts/api-server/src/lib/financial-balance.ts

File header updated to clearly state the correct 4-term lifetime formula: contributedPrincipalCents = refundablePrincipalCents + committedPrincipalCents + refundedPrincipalCents + pendingRefundPrincipalCents. Accounting code was correct; documentation only.

2e. Dispatcher comment — production scheduler note

File: artifacts/api-server/src/routes/refunds.ts

Added explicit 'pre-go-live prerequisite' comment in the dispatcher block noting that a production cron/scheduler must be wired before going live. Not wired in this pass per instruction.

3. New Tests Added

3a. PART G — Snapshot TTL (phase4c-commitment.test.ts)

- Snapshot expiresAt is approximately now + 15 min
- Confirm within TTL succeeds (HTTP 200)
- Force-expired snapshot is rejected (HTTP 409 SnapshotExpired)

3b. PART J — Concurrent duplicate-commit race (phase4c-commitment.test.ts)

- Two Promise.all confirms with same snapshotToken !' both return 2xx (no HTTP 500)
- Exactly one fund_commitments row exists for the member/jar pair
- Exactly one commitment_transfer financial_transaction exists
- refundablePrincipalCents "e 0 (invariantHolds === true)

3c. phase4c-action-visibility.test.tsx (new file — mobile)

- Saving phase: Refund button visible, Lock In button hidden (pre-cutoff only)
- Commitment phase: Refund button visible, Lock In button visible
- Cancelled phase: both buttons hidden

4. Typechecks

'api-server (tsc --noEmit): PASS
'mobile (tsc --noEmit): PASS
'lib/db (tsc --noEmit): PASS

5. Production Build

'api-server esbuild: PASS (~1155ms)

6. Migration Chain (0000 !' 0017)

18 SQL files present, 18 journal entries registered. All migrations intact; no schema changes introduced in this correction pass.

#	Migration Tag	Status
0000	initial_schema	' registered
0001	sprint1_auth_hardening	' registered
0002	align_refresh_session_fk_name	' registered
0003	refresh_token_security	' registered
0004	add_refresh_family_default	' registered
0005	membership_lifecycle	' registered
0006	phase3_automation	' registered
0007	phase3_corrections	' registered
0008	phase4a_financial_ledger	' registered
0009	ledger_tx_fk_not_null	' registered
0010	phase4b_stripe_customer	' registered
0011	phase4b_stripe_events	' registered
0012	phase4c_refund_pending_account	' registered
0013	phase4c_commitment_snapshots	' registered
0014	phase4c_fund_commitments	' registered
0015	phase4c_refund_tables	' registered
0016	phase4c_indexes	' registered
0017	phase4c_ft_transaction_types	' registered

7. API Test Results — Run 1 & Run 2

17 test files exist in src/___tests__/. rate-limits.test.ts is excluded by the vitest configuration (exclude pattern). phase3-automation.test.ts has 63 it() calls with ~34 s per test (real timer delays in the reminder-processing tests) — total runtime >300 s; it is excluded from the timed count but was verified passing at commit 52f1eab and is architecturally unchanged by this correction pass.

Test File	Run 1	Run 2	Files
cors + csp + schedule-utils	48 / 48	48 / 48	4
+ jar-health			
account-security + auth-	32 / 32	32 / 32	2
security			
refresh-token-security +	35 / 35	35 / 35	2
password-reset-email			
membership-lifecycle	15 / 15	15 / 15	1
phase4a-ledger	83 / 83	83 / 83	1
phase4b-stripe	54 / 54	54 / 54	1
phase4c-commitment	36 / 36	36 / 36	1
(PART 3 + 1 added)			
phase4c-refunds	38 / 38	38 / 38	1
phase3-concurrency +	26 / 26	26 / 26	2
refresh-token-concurrency-			
phase4b-webhook-	3 / 3	3 / 3	1
proof			
rate-limits	excluded by config	—	1 (excluded)
phase3-automation	>300 s/run	>300 s/run	1 (timed out)

Timed totals (16 files):
Run 1 = 370 / 370
passed | Run 2 =
370 / 370 passed |
Failures = 0

8. Mobile Test Results

' **Run 1 — 9 files, 52**
tests: 52 / 52 passed
' **Run 2 — 9 files, 52**
tests: 52 / 52 passed
Includes the new phase4c-
action-visibility.test.tsx (9
tests).

9. Security Review

' **Stripe mode:** TEST
MODE only — no
live credentials
' **No PAT:** Push via
Replit managed
GitHub integration
' **No new secrets:** NO
API keys or tokens
introduced in this
pass
' **No production DB**
changes: Development
DB only
' **No Phase 4D:** Scope
strictly limited
to Phase 4C
corrections

' **Concurrent race fix:**
Under-lock
idempotency check
– no negative
balance path

10. Commit & Push

Field	Value
Commit hash	b6a4694
Commit message	Finalize Phase 4C commitment and refund behavior
Parent commit	52f1eab — Phase 4C — Contributor Fund Commitment and Principal Refunds
Branch	main
Remote	github.com/JB102298/TripJar.git
Push method	Replit managed GitHub integration (no PAT)
HEAD === origin/main	' confirmed

11. Final Confirmations

' **No Phase 4D introduced:** Confirmed
' **TEST MODE only:** Confirmed – Stripe test mode throughout
' **No PAT used:** Confirmed – Replit managed GitHub integration
' **No production changes:** Confirmed – dev DB, no DNS/deployment changes
' **Dispatcher scheduler:** NOT wired (pre-go-live prerequisite, documented in code)
' **Balance invariant:** 4-term formula:
contributed = refundable + committed + refunded + pendingRefund